Software list

DFU Thermography Screening Station (DFUSCAN-01) · eMathrix Technologies · October 28, 2026

Every piece of software the station uses, where it runs, and what it is for. The station runs fully offline. No part of it sends scans, images or session data anywhere.

1. On the station (Raspberry Pi 5)

Operating system and runtime

Software	Version	Purpose
Raspberry Pi OS, 64-bit (desktop)	Bookworm or later	Operating system
Python	3.11 or later	Language of all station software
Python virtual environment	created with `--system-site-packages`	Lets the app use the system camera and Hailo packages, which must not be reinstalled through pip

System packages (installed with apt)

Package	Purpose
`python3-picamera2` (libcamera)	Camera Module 3 colour frames, with focus, exposure and white balance locked
`hailo-all` (HailoRT, `hailo_platform`)	Runs the compiled classifier (`.hef`) on the AI HAT+ (Hailo-8L)
`hailo-models`	YOLOv8s object detector used only for the live preview boxes in the foot photo app. It is not a foot or wound model.
`python3-lgpio`	GPIO driver for the Pi 5 (RP1 chip). RPi.GPIO does not work on the Pi 5 and is not used.
`python3-gpiozero`	Capture button, status LED and interior-light switch
`python3-pyqt5`	Full-screen touchscreen interface of the station app
`python3-flask` , `python3-pil`	Foot photo capture app (local web server and image handling)
`chromium` , `wlrctl` , `curl`	Shows the foot photo app full-screen on the touchscreen (kiosk mode)

Python packages (installed with pip)

Package	Version	Purpose
NumPy	1.26 or later	Temperature arrays and image maths
OpenCV (headless)	4.8 or later	Registration, segmentation, image processing
PyYAML	6.0 or later	Station configuration and calibration files
gpiozero	2.0 or later	GPIO devices (same as above, for laptop testing)
smbus2	0.4 or later	SHT31 ambient sensor over I^2C

Station applications

Application	Started by	What it does

Screening station app	`python -m src.app` (systemd service `dfu-station`)	The full scan: ambient and acclimatisation checks, paired colour and thermal capture, registration, left/right asymmetry with the 2.2 °C flag, classifier, offline storage and CSV
Foot photo capture app	`python -m capture_app` (desktop icon or autostart)	Colour photos for data gathering before a model exists, with USB export
Hardware self-check and calibration tools	`python -m src.diagnostics.…`	Checks each device; thermal, colour-lock and registration calibration

Thermal camera interface

Item	Purpose
Linux V4L2 / UVC (built into the OS)	Reads the PureThermal 3 as a USB camera, Y16 radiometric stream
GroupGets `purethermal1-uvc-capture` (reference)	Reference for Lepton control over UVC: flat-field correction and radiometry (TLinear)

2. Training machine (x86-64 PC, never the Pi)

Software	Version	Purpose
Python	3.11 or later	Training scripts in `training/`
PyTorch	2.5 or later	Training the classifier
torchvision	0.20 or later	MobileNetV2 network with ImageNet starting weights
ONNX	1.16 or later	Exported model format
ONNX Runtime	1.18 or later	Checks the export matches PyTorch; laptop testing of the model
Hailo Dataflow Compiler (`hailo_sdk_client`)	from the Hailo Developer Zone	Compiles the ONNX model to `.hef` for the Hailo-8L. Needs x86-64 Linux with 16 GB RAM or more.

3. Development and testing (laptop)

Software	Purpose
pytest 8 or later	Automated tests. The whole pipeline runs on a laptop with fake hardware.
ruff, mypy, types-PyYAML	Code style and type checks
PyQt5, ONNX Runtime	Laptop stand-ins for the Pi's touchscreen app and Hailo runner
Git, GitHub	Source code. Scans, calibration files and session data are never committed.

4. Documents page

Software	Purpose
GitHub Pages and Railway (Python 3.11, Markdown 3.7)	Public page with manuals, schematic, bill of materials and the 3D model. It holds no patient data and has no upload.
three.js (in the browser)	Interactive 3D model of the station

5. Models

Model	Used for	Notes
MobileNetV2 (trained by the project)	Classifier on the registered colour image, on the Hailo-8L	Outputs a predicted class (`class_0` , `class_1` , …), never a Wagner grade. See the training document.
YOLOv8s (ships with Raspberry Pi OS)	Preview boxes in the foot photo app only	General object detector; never written into saved photos

> **Not a model:** the left–right temperature asymmetry and its 2.2 °C flag are plain arithmetic on calibrated temperatures. They are deterministic and are not learned.